Security Control of Vehicle Platoons under Hybrid Attacks via Meta-Gradient Reward Shaping

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***Abstract*—Vehicle platooning has become an important technology for improving traffic efficiency, safety, and coordinated mobility in intelligent transportation systems. Vehicle platoons rely on vehicle-to-everything (V2X) communication, exposing uplink observations and downlink control commands to denial-of-service (DoS) and false data injection (FDI) attacks. When reinforcement learning is employed for vehicle platoon control, hybrid DoS–FDI attacks can corrupt state information, physical actuation, and environmental rewards derived from communicated observations. Such corruption may cause control decisions and policy updates to deviate from the actual platoon-control objective. To address this issue, a hierarchical Meta-Gradient Reward Shaping (MGRS) framework is proposed for security control of vehicle platoons under hybrid attacks. A Conditional Attack Generative Adversarial Network (CA-GAN) generates state-conditioned hybrid attacks. Lower-level Heterogeneous-Agent Trust-Region Learning (HATRL) student policies learn decentralized longitudinal control from attacked local observations. An upper-level physics-aware Reward Teacher adaptively adjusts the reward-shaping term using uncompromised physical states and executed inputs available during centralized training. Furthermore, a first-order teacher meta-gradient is derived, and a conditional local ultimate-boundedness result is established. Simulation results demonstrate favorable learning and control performance under hybrid attacks, with a high cumulative return, low speed and spacing errors, and a small centerline geometric deviation. Physical experiments further validate the feasibility of the proposed method, with centimeter-scale radial geometric deviations.**

***Index Terms*—Vehicle platooning, meta-gradient reward shaping, reinforcement learning, hybrid attacks.**

I. Introduction

With the rapid development of autonomous driving and vehicle-to-everything (V2X) communication, connected vehicle platooning has become an important technology for intelligent transportation systems [1], [2]. Cooperative adaptive cruise control (CACC) coordinates longitudinal vehicle motion through inter-vehicle information exchange, enabling safe spacing and smooth car-following while improving traffic throughput, energy efficiency, and ride comfort [3], [4]. By exploiting V2X information beyond onboard sensing, CACC enables cooperative decision-making but also makes reliable platoon coordination increasingly dependent on communication integrity and availability [3], [5].

Recent studies have further investigated communication-aware and data-driven vehicle coordination. Platoon stability and transient inter-vehicle distances under bidirectional communication topologies are analyzed in [6], showing that internal stability alone does not necessarily guarantee safe transient spacing. A data-enabled predictive leading cruise control framework for constrained vehicle coordination is developed in [7] without relying on an explicit parametric car-following model. However, the openness of wireless V2X networks exposes vehicle platoons to denial-of-service (DoS) and false data injection (FDI) attacks [8]–[11]. DoS attacks cause packet losses or communication delays, forcing controllers to operate with missing or outdated information, whereas FDI attacks manipulate transmitted vehicle states or control commands [12]–[14]. Coordinated DoS–FDI attacks may further produce dynamically varying packet-loss and data-injection patterns, resulting in degraded platoon-control performance [15]. Therefore, robustness to random noise, fixed disturbances, or predefined attack patterns is insufficient against adaptive hybrid DoS–FDI network attacks.

To mitigate the effects of such cyberattacks, various security-control strategies have been developed. Security resource allocation in networked control systems is formulated as a robust Stackelberg game in [16], while centralized and distributed mechanisms for cyberattack and machine-fault detection and isolation are developed in [17]. Communication-security issues in decentralized learning under adversarial eavesdropping are investigated in [18]. For vehicle platoons, fault-tolerant control has been studied under aperiodic DoS attacks and sensor/actuator faults [19], as well as under actuator faults with dead-zone and saturation nonlinearities [20]. A hierarchical game-based security-control framework against FDI attacks is further proposed in [21] by combining differential graph and zero-sum dynamic games. These approaches mainly rely on model-based estimation, controller synthesis, or explicit optimization, which can become restrictive when vehicle interactions and attack behaviors are highly nonlinear and dynamically varying. Therefore, multi-agent reinforcement learning (MARL) has increasingly been applied to adaptive security control of vehicle platoons. Under centralized training with decentralized execution (CTDE), global information can be

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exploited during training while each vehicle executes its policy from local observations [22]. Representative methods include Multi-Agent Proximal Policy Optimization (MAPPO) [23] and heterogeneous-agent algorithms such as Heterogeneous-Agent Proximal Policy Optimization (HAPPO) and Heterogeneous-Agent Trust Region Policy Optimization (HATRPO) [24], [25], which improve cooperative policy optimization through sequential policy updates. However, under hybrid DoS–FDI attacks, corrupted observations can affect not only action selection but also reward evaluation.

When environmental rewards are computed from falsified, delayed, or held vehicle information, they may no longer reflect the actual tracking and motion-smoothness performance of the platoon, resulting in reward misalignment. Consequently, policy updates may be driven toward physically undesirable actions even when the learning process remains numerically stable. Therefore, reinforcement-learning-based security control of vehicle platoons requires not only adaptation to corrupted observations but also correction of attack-induced reward misalignment.

Recent studies have explored learned reward generation and bi-level reward discovery to reduce dependence on handcrafted rewards [26], [27], while meta-gradient reinforcement learning adapts learning objectives through an outer performance criterion [28], [29]. However, these approaches do not address reward misalignment caused by simultaneous observation and actuation attacks in vehicle platoons. Safe reinforcement learning mainly constrains unsafe actions or policy updates [30], [31], but such constraints do not directly correct an attack-distorted reward signal.

To address these limitations, a hierarchical Meta-Gradient Reward Shaping (MGRS) framework is proposed for security control of vehicle platoons under hybrid attacks, integrating an adversarial environment based on a Conditional Attack Generative Adversarial Network (CA-GAN), lower-level Heterogeneous-Agent Trust-Region Learning (HATRL) students [32], and an upper-level physics-aware Reward Teacher. The CA-GAN generates state-conditioned hybrid DoS–FDI attacks, while the HATRL students learn decentralized longitudinal control from attacked local observations. Using uncompromised physical states and executed inputs available during centralized training, the Reward Teacher evaluates post-update student performance and adapts the shaping term to correct attack-induced reward misalignment; a training-time action projection is additionally used to limit unsafe exploration.

The main contributions of this work are summarized as follows.

1) A hierarchical MGRS framework is proposed to address attack-induced reward misalignment in vehicle-platoon security control. By using uncompromised physical-performance information available during centralized training to adapt the reward-shaping signal, the learned policies are guided toward the actual platoon-control objective despite corrupted observations and control commands.

2) A state-conditioned CA-GAN is developed to generate hybrid DoS–FDI attacks jointly on uplink observations and downlink control commands under explicit attack-budget constraints. The generated attacks vary with the current platoon state, enabling the learned policies to adapt to dynamically changing hybrid attacks.

3) Student adaptation and reward adjustment are formulated within a bi-level learning framework. A first-order Teacher meta-gradient is derived by using a differentiable one-step surrogate for the practical finite-step HATRL update, and conditional local ultimate boundedness is established for the deployed platoon under bounded attacks, external disturbances, and closed-loop approximation residuals.

The remainder of this paper is organized as follows. Section II introduces the platoon dynamics, hybrid DoS–FDI attack model, and control objectives. Section III presents the hierarchical MGRS framework, and Section IV provides the teacher meta-gradient and conditional local ultimate-boundedness analyses. Sections V and VI present the simulation and physical-platform results, respectively, followed by the conclusion in Section VII.

II. Preliminaries and Problem Formulation

This section formulates the platoon dynamics, the hybrid DoS–FDI attack model, and the corresponding physical performance criteria.

A. Vehicle Platoon Dynamics and V2X Communication

Consider a vehicle platoon consisting of one leader and m follower vehicles. The longitudinal dynamics of the i-th follower are described by the following third-order nonlinear model:

$$\begin{cases} \dot{p}_i(t) = v_i(t), \\ \dot{v}_i(t) = a_i(t), \\ \dot{a}_i(t) = f_i(v_i(t), a_i(t)) + \dfrac{1}{\tau_i} u_{i,\mathrm{act}}(t) + w_i(t), \\ y_i(t) = C_i X_i(t) + \delta_i(t), \end{cases} \tag{1}$$

where $X_i(t) = [p_i(t), v_i(t), a_i(t)]^\top$ is the vehicle state, with $p_i(t)$, $v_i(t)$, and $a_i(t)$ denoting the longitudinal position along the prescribed reference path, velocity, and acceleration, respectively. The nonlinear term $f_i(v_i(t), a_i(t))$ captures inherent vehicle dynamics such as aerodynamic drag and mechanical resistance; τ_i is the driveline inertial time constant; $w_i(t)$ represents external disturbances; and $y_i(t)$ is the measured output with $C_i = I_3$ and measurement noise $\delta_i(t)$. The signal $u_{i,\mathrm{act}}(t)$ denotes the control input physically applied to the i-th vehicle.

The physical measurement y_i is distinct from the local feature observation used by the learning interface. For each vehicle,

$$\begin{aligned} o_i[n] &= \mathrm{col}\left\{ \begin{array}{c} \Delta p_{x,i}, \Delta p_{y,i}, \Delta v_{x,i}, \Delta v_{y,i}, \\ v_{x,i}, v_{y,i}, \omega_{z,i}, e_{c,i}, \\ \sin(e_{\psi,i}), \sin(\Delta\psi_i), \varrho_i \end{array} \right\}[n], \\ o_i[n] &\in \mathbb{R}^O, \qquad O = 11. \end{aligned} \tag{2}$$

where $(\Delta p_{x,i}, \Delta p_{y,i})$ and $(\Delta v_{x,i}, \Delta v_{y,i})$ denote the predecessor-relative position and velocity differences in the body frame; $(v_{x,i}, v_{y,i})$ are the ego velocity components; $\omega_{z,i}$ is the yaw rate; $e_{c,i}$ and $e_{\psi,i}$ are the centerline

and heading errors relative to the reference path; $\Delta\psi_i$ is the predecessor-relative heading difference; and ϱ_i is the normalized vehicle-role identifier. Although the control input is longitudinal, these planar geometric features characterize the local vehicle configuration relative to the predecessor and reference path. The same 11-dimensional structure is used for the leader, with unavailable predecessor-related entries set to zero.

The platoon adopts a cloud-assisted V2X architecture in which local observations and control commands are exchanged over wireless uplink and downlink channels, exposing both channels to hybrid DoS–FDI network attacks.

Mathematically, the hybrid DoS–FDI attack model for the i-th vehicle is expressed as:

$$\begin{aligned}\tilde{u}_i[n] &= \beta_{a,i}[n] \odot \big[u_i[n] + f_{a,i}[n]\big] \\ &\quad + \big(\mathbf{1}_U - \beta_{a,i}[n]\big) \odot u_{i,\mathrm{hold}}[n], \\ \tilde{o}_i[n] &= \beta_{p,i}[n] \odot \big[o_i[n] + f_{p,i}[n]\big] \\ &\quad + \big(\mathbf{1}_O - \beta_{p,i}[n]\big) \odot o_{i,\mathrm{hold}}[n].\end{aligned} \tag{3}$$

where $\odot$ denotes the Hadamard product, $U = 1$ is the longitudinal-control dimension, and $O = 11$ is the local-observation dimension. The vectors $\mathbf{1}_U$ and $\mathbf{1}_O$ denote all-one vectors with dimensions U and O, respectively. The vectors $\beta_{a,i} \in \{0,1\}^U$ and $\beta_{p,i} \in \{0,1\}^O$ are component-level transmission masks, while $f_{a,i} \in \mathbb{R}^U$ and $f_{p,i} \in \mathbb{R}^O$ denote the control- and observation-channel FDI perturbations. For dropped components, $u_{i,\mathrm{hold}}$ and $o_{i,\mathrm{hold}}$ retain the most recent valid values through zero-order hold (ZOH).

The physical vehicles evolve continuously according to Eq. (1), whereas student actions, hybrid DoS–FDI attacks, and the training-time action projection are evaluated at the sampling instants

$$t_n = nT_s, \qquad n = 0, 1, \ldots, \tag{4}$$

where $T_s > 0$ is the control sampling period and $X_i[n] := X_i(t_n)$.

Under the standard sample-and-hold implementation, the executed input is held constant over each control interval as

$$u_{i,\mathrm{act}}(t) = u_{i,\mathrm{act}}[n], \qquad t \in [t_n, t_{n+1}), \tag{5}$$

yielding the sampled dynamics

$$X_i[n+1] = \mathcal{F}_{i,T_s}\left(X_i[n], u_{i,\mathrm{act}}[n], w_i[n]\right), \tag{6}$$

where $\mathcal{F}_{i,T_s}$ is the one-step transition mapping obtained by integrating Eq. (1) over one sampling interval, and $w_i[n]$ denotes the corresponding sampled disturbance. The index n is used hereafter for attack realizations, control actions, and one-step predictions. The physically executed input is

$$u_{i,\mathrm{act}}[n] = \begin{cases} \hat{u}_i[n], & \text{during centralized adversarial training,} \\ \tilde{u}_i[n], & \text{during deployment without the training-time projection,} \end{cases} \tag{7}$$

where $\hat{u}_i[n]$ denotes the projected training-time input defined in Section III-B.

B. CA-GAN-Based Hybrid DoS–FDI Attack Model

The CA-GAN generates state-conditioned hybrid DoS–FDI attacks through a generator G and a discriminator D. Conditioned on the clean local-feature observation $o_i[n]$ and a random noise vector $z_i[n]$, the generator produces the FDI perturbations and component-level DoS dropout probabilities,

$$\xi_i[n] = G_{\theta_G}(z_i[n], o_i[n]) = [f_{p,i}, q_{p,i}, f_{a,i}, q_{a,i}]\,[n]. \tag{8}$$

Here, θ_G denotes the generator parameters. Binary transmission masks are subsequently sampled from the DoS probabilities as specified in Eq. (10).

Mechanism-consistent reference attacks are constructed from typical DoS–FDI behaviors and admissible attack constraints. The discriminator distinguishes generated attacks from these reference attack samples, thereby providing attack-realism regularization toward feasible hybrid attack patterns. The corresponding adversarial regularization term for the generator is defined as

$$\mathcal{L}_{\mathrm{real}}(\theta_G; D) = \mathbb{E}_{o_i \sim p_o,\, z_i \sim p_z}\left[-\log D(G_{\theta_G}(z_i, o_i) \mid o_i)\right]. \tag{9}$$

Here, p_o and p_z denote the clean local-feature and noise distributions, respectively.

To ensure that the generated attacks remain bounded and satisfy the prescribed attack budgets, the CA-GAN heads are parameterized as

$$\begin{aligned} &f_{p,i}[n] = \varepsilon_p \tanh\big(\eta_{p,i}[n]\big), && \|f_{p,i}[n]\|_\infty \le \varepsilon_p, \\ &f_{a,i}[n] = \varepsilon_a \tanh\big(\eta_{a,i}[n]\big), && \|f_{a,i}[n]\|_\infty \le \varepsilon_a, \\ &q_{p,i}^{(j)}[n] = \rho_{\mathrm{DoS}}^{\max}\sigma\big(\ell_{p,i}^{(j)}[n]\big), \\ &d_{p,i}^{(j)}[n] \sim \mathrm{Bernoulli}\big(q_{p,i}^{(j)}[n]\big), \\ &q_{a,i}^{(k)}[n] = \rho_{\mathrm{DoS}}^{\max}\sigma\big(\ell_{a,i}^{(k)}[n]\big), \\ &d_{a,i}^{(k)}[n] \sim \mathrm{Bernoulli}\big(q_{a,i}^{(k)}[n]\big), \\ &\beta_{p,i}[n] = \mathbf{1}_O - d_{p,i}[n], && \beta_{a,i}[n] = \mathbf{1}_U - d_{a,i}[n]. \end{aligned} \tag{10}$$

where $\eta_{p,i}$ and $\eta_{a,i}$ are the FDI-output heads, $\ell_{p,i}^{(j)}$ and $\ell_{a,i}^{(k)}$ are the DoS logits, and $\sigma(\cdot)$ is the sigmoid function, with $j = 1, \ldots, O$ and $k = 1, \ldots, U$. The resulting dropout probabilities satisfy $0 \le q_{p,i}^{(j)}, q_{a,i}^{(k)} \le \rho_{\mathrm{DoS}}^{\max}$, where $\rho_{\mathrm{DoS}}^{\max}$ is the shared component-level DoS-rate bound. The parameters ε_p and ε_a bound the observation- and control-channel FDI perturbations, respectively. A bounded consecutive-loss length is additionally imposed to keep the packet-loss-induced holding error bounded. Here, $d_{p,i}$ and $d_{a,i}$ denote binary dropout indicators sampled from the corresponding DoS probabilities.

C. Control Objective and Physical Performance Criteria

The control objective is to maintain platoon coordination and keep the spacing, velocity, and acceleration tracking errors within prescribed ultimate bounds under admissible hybrid DoS–FDI network attacks and external disturbances. Let the leader be indexed by 0 and the followers by $i = 1, \ldots, m$.

Under the constant time headway (CTH) policy, the sampled tracking errors are defined as

$$\begin{aligned} e_{p,i}[n] &= p_{i-1}[n] - p_i[n] - d_0 - hv_i[n], \\ e_{v,i}[n] &= v_{i-1}[n] - v_i[n], \\ e_{a,i}[n] &= a_{i-1}[n] - a_i[n]. \end{aligned} \tag{11}$$

where $d_0 > 0$ and $h > 0$ denote the standstill spacing and prescribed time headway, respectively. Because persistent attacks and disturbances may prevent exact convergence, the tracking objective is formulated as

$$\limsup_{n\to\infty} |e_{q,i}[n]| \le \delta_q, \qquad q \in \{p, v, a\}, \quad i = 1, \dots, m. \tag{12}$$

The index $q \in \{p, v, a\}$ denotes the error type associated with spacing, velocity, and acceleration, respectively. Here, $\delta_p, \delta_v, \delta_a > 0$ are the prescribed ultimate margins for the spacing, velocity, and acceleration errors, respectively. Section IV establishes sufficient local conditions under which Eq. (12) is satisfied.

To quantify physical degradation and adversarial expenditure, define

$$c_i[n] = \zeta\Big(l_{\mathrm{perf},i}[n] - l_{\mathrm{adv},i}[n]\Big), \tag{13}$$

where $\zeta > 0$ is a normalization factor, and

$$\begin{aligned} l_{\mathrm{perf},i}[n] &= \lambda_p \|e_{p,i}[n]\|^2 + \lambda_v \|e_{v,i}[n]\|^2 \\ &\quad + \lambda_a \|e_{a,i}[n]\|^2 + \lambda_{\mathrm{jerk}} \left\| \frac{a_i[n] - a_i[n-1]}{T_s} \right\|^2 \\ &\quad + \lambda_{\mathrm{over}} \left[\max\big(0, v_i[n] - v_{\mathrm{tar}}[n]\big)\right]^2, \\ l_{\mathrm{adv},i}[n] &= \nu \|f_{a,i}[n]\|^2 + \mu \|f_{p,i}[n]\|^2 \\ &\quad + \frac{\kappa_1}{U} \mathbf{1}_U^\top d_{a,i}[n] + \frac{\kappa_2}{O} \mathbf{1}_O^\top d_{p,i}[n]. \end{aligned} \tag{14}$$

Here, λ_p, λ_v, and λ_a are weighting coefficients for the spacing, velocity, and acceleration tracking errors, respectively; λ_{jerk} and λ_{over} denote the penalty coefficients for jerk and overspeed. The desired velocity reference at sampling instant n is denoted by $v_{\mathrm{tar}}[n]$. Moreover, ν and μ weight the control- and observation-channel FDI magnitudes, while κ_1 and κ_2 weight the corresponding DoS rates. Therefore, $l_{\mathrm{perf},i}$ quantifies physical performance degradation and $l_{\mathrm{adv},i}$ represents adversarial expenditure. The platoon-level degradation objective is

$$J(\theta, \theta_G) = \mathbb{E}\left[\sum_{n=0}^{H-1} \gamma^n \left(\frac{1}{m}\sum_{i=1}^{m} c_i[n] + C_{\mathrm{col}} \mathbb{I}_{\mathrm{col}}[n]\right)\right], \tag{15}$$

where H denotes the rollout horizon and $\gamma \in (0, 1)$ is the discount factor. Moreover, $C_{\mathrm{col}} > 0$ is the collision penalty coefficient and $\mathbb{I}_{\mathrm{col}}[n]$ is the collision indicator. The CA-GAN objective is

$$\mathcal{J}_{\mathrm{att}}(\theta, \theta_G) = J(\theta, \theta_G) - \lambda_{\mathrm{real}} \mathcal{L}_{\mathrm{real}}(\theta_G; D), \tag{16}$$

where $\lambda_{\mathrm{real}} > 0$ balances physical degradation and attack-realism regularization. During generator optimization, the continuous DoS-probability surrogate is used, while binary Bernoulli masks are sampled for adversarial rollouts.

III. Hierarchical Meta-Gradient Reward Shaping Framework

This section develops the hierarchical MGRS framework in the CA-GAN-based adversarial environment, comprising lower-level HATRL student adaptation, an upper-level physics-aware Reward Teacher, and a training-time action projection.

A. Bi-Level Student–Teacher Architecture

The lower level updates decentralized student policies using the shaped reward, while the upper level adapts the reward-shaping function according to post-update physical performance, as illustrated in Fig. 1.

1) Lower-Level: Student Policy Optimization via HATRL: For each follower i, the student policy $\pi_{\theta_i}(u_i[n] \mid \tilde{o}_i[n])$ generates the longitudinal action $u_i[n]$ from the attacked local observation $\tilde{o}_i[n]$, with $\theta = \mathrm{col}\{\theta_1, \dots, \theta_m\}$. Under CTDE, the centralized critic receives the corrupted joint observation

$$\tilde{s}[n] = \mathrm{col}\{\tilde{o}_0[n], \dots, \tilde{o}_m[n]\}. \tag{17}$$

For reward evaluation, the attack-corrupted spacing error is defined as

$$\tilde{e}_{p,i}[n] = \tilde{d}_i[n] - d_0 - h\tilde{v}_i[n], \tag{18}$$

where $\tilde{d}_i[n]$ and $\Delta\tilde{v}_i[n]$ denote the attacked longitudinal gap and relative-velocity features, respectively, with $\tilde{e}_{v,i}[n] = \Delta\tilde{v}_i[n]$.

$$r_i^{\mathrm{env}}[n] = -\tilde{l}_{\mathrm{env},i}[n], \tag{19}$$

where $\tilde{l}_{\mathrm{env},i}[n]$ is the attack-corrupted counterpart of the physical-performance loss in Eq. (14). The shaped reward is

$$R_\phi\left(\tilde{o}_i[n], u_i[n]\right) = r_i^{\mathrm{env}}[n] + F_\phi\left(\tilde{o}_i[n], u_i[n]\right). \tag{20}$$

Accordingly, the lower-level objective is

$$J_{\mathrm{in}}(\theta; \phi) = \mathbb{E}\left[\sum_{n=0}^{H-1} \gamma^n \frac{1}{m} \sum_{i=1}^{m} R_\phi\left(\tilde{o}_i[n], u_i[n]\right)\right], \tag{21}$$

with the ideal optimizer satisfying

$$\theta^\star(\phi) \in \arg\max_\theta J_{\mathrm{in}}(\theta; \phi). \tag{22}$$

In practice, the ideal optimizer $\theta^\star(\phi)$ is approximated by the output of the finite-step HATRL update

$$\theta^+ = \mathcal{U}_{\mathrm{HATRL}}(\theta^-, \phi; \mathcal{B}). \tag{23}$$

2) Upper-Level: Physics-Aware Reward Teacher: The upper-level physical criterion is defined by

$$l_{\mathrm{proj}}[n] = \frac{\lambda_{\mathrm{proj}}}{m} \sum_{i=1}^{m} \|\hat{u}_i[n] - \tilde{u}_i[n]\|^2, \qquad \lambda_{\mathrm{proj}} \ge 0, \tag{24}$$

and

$$l_{\mathrm{phy}}[n] = \frac{1}{m} \sum_{i=1}^{m} l_{\mathrm{perf},i}^{\mathrm{true}}[n] + C_{\mathrm{col}} \mathbb{I}_{\mathrm{col}}[n] + l_{\mathrm{proj}}[n], \tag{25}$$

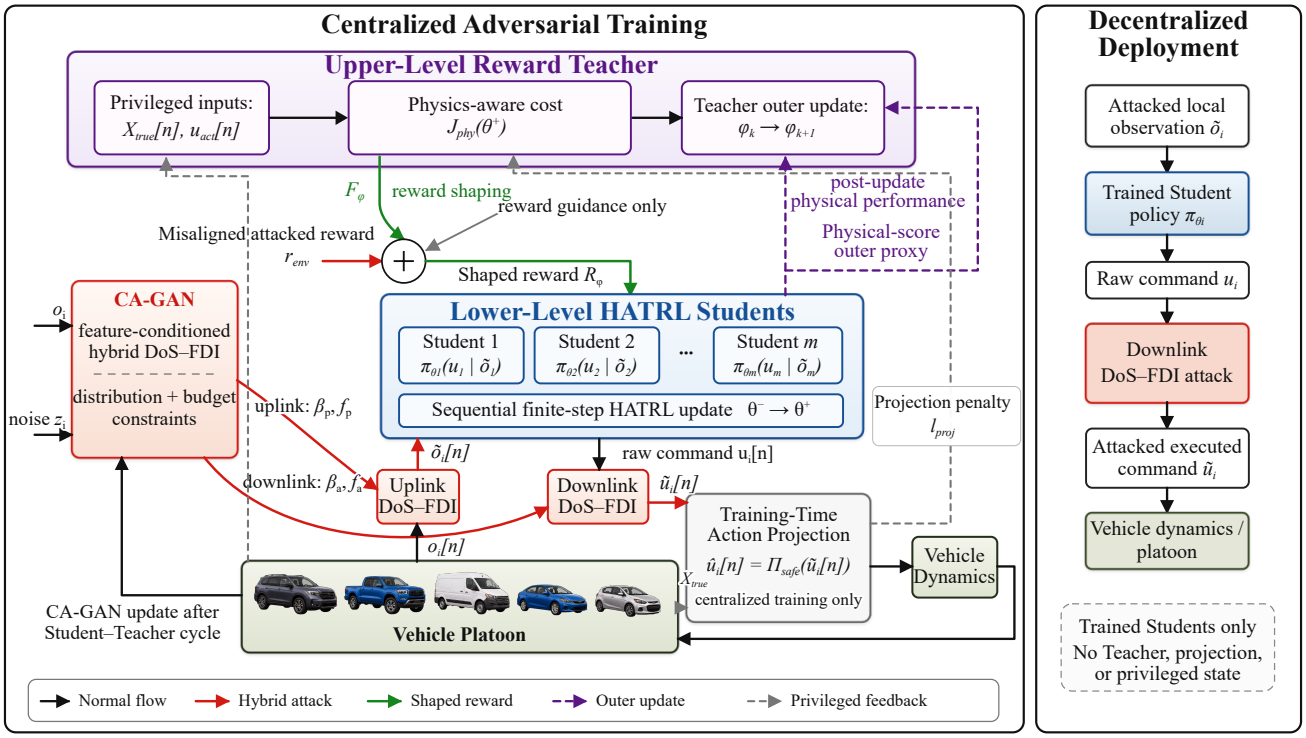


Fig. 1. Architecture of the proposed MGRS framework for centralized adversarial training and decentralized deployment.

where $l_{\mathrm{perf},i}^{\mathrm{true}}$ follows Eq. (14) using uncompromised physical quantities, and λ_{proj} weights the projection intervention. The corresponding physical objective is

$$J_{\mathrm{phy}}(\theta) = \mathbb{E}\left[\sum_{n=0}^{H-1} \gamma^n l_{\mathrm{phy}}[n]\right]. \tag{26}$$

The upper-level objective and the resulting bi-level problem are

$$J_{\mathrm{out}}(\phi) := J_{\mathrm{phy}}\left(\theta^\star(\phi)\right), \tag{27}$$

and

$$\min_{\phi} \; J_{\mathrm{out}}(\phi) \quad \text{s.t.} \quad \theta^\star(\phi) \in \arg\max_{\theta} J_{\mathrm{in}}(\theta;\phi). \tag{28}$$

In practical training, $\theta^\star(\phi)$ is replaced by the finite-step HATRL output θ^+, and the teacher is updated as

$$\phi_{k+1} = \phi_k - \beta_k \widehat{g}_{\phi,k}^{\mathrm{outer}}, \tag{29}$$

where $\widehat{g}_{\phi,k}^{\mathrm{outer}}$ is the first-order outer direction specified in Section IV-A.

B. Training-Time Action Projection

To limit unsafe exploration during centralized adversarial training, the attacked command $\tilde{u}_i[n]$ is processed by

$$u_i[n] \longrightarrow \tilde{u}_i[n] \longrightarrow \hat{u}_i[n]. \tag{30}$$

Let the inter-vehicle gap and its one-step prediction under candidate input u be defined as

$$\begin{aligned} d_i[n] &= p_{i-1}[n] - p_i[n], \\ d_i^+[n;u] &= \mathcal{F}_{d,i,T_s}\left(X_{\mathrm{true}}[n], u_{i-1,\mathrm{act}}[n], u\right). \end{aligned} \tag{31}$$

Here, $\mathcal{F}_{d,i,T_s}$ denotes the discrete gap-transition mapping obtained from the vehicle dynamics with sampling period T_s.

The projected input is

$$\hat{u}_i[n] = \Pi_{\mathrm{safe}}\left(\tilde{u}_i[n]\right) = \begin{cases} u_{\mathrm{brake}}^{\max}, & d_i^+\left[n;\tilde{u}_i[n]\right] \le d_{\mathrm{crit}}, \\ u_{\mathrm{catch}}, & d_i[n] \ge d_{\mathrm{drop}}, \\ & v_i[n] \le v_{i-1}[n] + \Delta v, \\ & d_i^+\left[n;u_{\mathrm{catch}}\right] > d_{\mathrm{crit}}, \\ \tilde{u}_i[n], & \text{otherwise}, \end{cases} \tag{32}$$

where $u_{\mathrm{brake}}^{\max}$, u_{catch}, d_{crit}, d_{drop}, and Δv denote the maximum braking command, the bounded catch-up command, the critical-gap threshold, the catch-up threshold, and the relative-speed margin, respectively. The projection is used only during centralized training and is removed during decentralized deployment.

IV. Meta-Gradient and Ultimate-Boundedness Analysis

This section derives a first-order teacher meta-gradient through a differentiable surrogate of the finite-step HATRL adaptation and establishes conditional local ultimate boundedness of the deployed platoon tracking errors.

A. Physics-Aware Teacher Meta-Gradient

Let θ^- denote the pre-update student parameter. The finite-step HATRL adaptation is

$$\theta^+ = \mathcal{U}_{\mathrm{HATRL}}\left(\theta^-, \phi; \mathcal{B}\right), \tag{33}$$

where $\mathcal{B}$ is the sampled trajectory batch.

Algorithm 1 Bi-Level MGRS Training under CA-GAN Attacks

Require: Sampled platoon dynamics, rollout horizon H, meta-training iterations K, student and teacher learning rates, and outer-proxy clipping and ramp parameters.

Ensure: Trained decentralized student policies θ.

1: Initialize the student policies θ, reward teacher ϕ, shaped-reward and physical-cost critics, and the CA-GAN generator and discriminator.
2: **for** meta-training iteration $k = 0, 1, \ldots, K-1$ **do**
3: Set $\theta_k^- = \theta_k$, reset the environment, and hold the CA-GAN parameters fixed.
4: Collect an adversarial trajectory batch under CA-GAN-generated attacks and the training-time action projection.
5: Compute the shaped rewards and lower-level advantages, and apply finite-step sequential HATRL to obtain

$$\theta_k^+ = \mathcal{U}_{\mathrm{HATRL}}\left(\theta_k^-, \phi_k; \mathcal{B}_k\right).$$

6: At an active teacher-update boundary, evaluate the clean physical score and its normalized change from the preceding active boundary.
7: Form the physical-score/teacher-output covariance and evaluate the physical-score-guided outer direction $\widehat{g}_{\phi,k}^{\mathrm{outer}}$ with the prescribed clipping and ramp.
8: Update

$$\phi_{k+1} = \phi_k - \beta_k \widehat{g}_{\phi,k}^{\mathrm{outer}}, \qquad \theta_{k+1} = \theta_k^+.$$

9: Update the CA-GAN generator and discriminator after completing the student–teacher updates.
10: **end for**

For first-order differentiation, introduce the one-step surrogate

$$\bar{\theta}^+ = \theta^- + \alpha g_{\mathrm{in}}\left(\theta^-, \phi\right), \tag{34}$$

with step size $\alpha > 0$, where

$$g_{\mathrm{in}}\left(\theta^-, \phi\right) = \mathbb{E}_{\mu^-}\left[A_\phi^-\left(\tilde{s}, u\right) \nabla_{\theta^-} \log \pi_{\theta^-}\left(u \mid \tilde{s}\right)\right], \tag{35}$$

and

$$A_\phi^-\left(\tilde{s}, u\right) = Q_\phi^{\pi_{\theta^-}}\left(\tilde{s}, u\right) - V_\phi^{\pi_{\theta^-}}\left(\tilde{s}\right), \tag{36}$$

and $\mu^- := \mu_{\theta^-}$ is the pre-update state–action distribution. Here, Q_ϕ^π and V_ϕ^π denote the action-value and state-value functions, respectively.

The corresponding surrogate outer objective is

$$\widetilde{J}_{\mathrm{out}}\left(\phi; \theta^-\right) = J_{\mathrm{phy}}\left(\bar{\theta}^+(\phi)\right). \tag{37}$$

Assumption 1. $A_\phi^-(\tilde{s}, u)$ is continuously differentiable in ϕ, while π_θ and $J_{\mathrm{phy}}(\theta)$ are continuously differentiable in θ, with integrable derivatives permitting interchange of differentiation and expectation. During truncated differentiation, θ^- and μ^- are held fixed with respect to ϕ.

Theorem IV.1 (First-order truncated Teacher meta-gradient)**.** *Under Assumption 1,*

$$\nabla_\phi \widetilde{J}_{\mathrm{out}}\left(\phi; \theta^-\right) = \left(\frac{\partial \bar{\theta}^+}{\partial \phi}\right)^\top \nabla_{\bar{\theta}^+} J_{\mathrm{phy}}\left(\bar{\theta}^+\right), \tag{38}$$

where

$$\frac{\partial \bar{\theta}^+}{\partial \phi} = \alpha \mathbb{E}_{\mu^-}\left[\nabla_{\theta^-} \log \pi_{\theta^-}\left(u \mid \tilde{s}\right)\left(\nabla_\phi A_\phi^-\left(\tilde{s}, u\right)\right)^\top\right]. \tag{39}$$

Proof. From Eq. (34),

$$\frac{\partial \bar{\theta}^+}{\partial \phi} = \alpha \frac{\partial g_{\mathrm{in}}\left(\theta^-, \phi\right)}{\partial \phi}. \tag{40}$$

Under Assumption 1, differentiation can be moved inside the expectation, while the pre-update sampling distribution and policy score remain fixed. Differentiating the shaped-reward advantage in Eq. (35) therefore yields Eq. (39); applying the chain rule to Eq. (37) gives Eq. (38). □

The result applies to the one-step surrogate in Eq. (34), not the finite-step HATRL map in Eq. (33).

The physical consequence of the raw student action is represented by

$$u_{\mathrm{act}}[n] = \mathcal{T}\left(X_{\mathrm{true}}[n], u[n], \xi[n]\right), \tag{41}$$

where $\mathcal{T}$ incorporates the realized downlink attack and the training-time action projection.

For the surrogate post-update policy, define

$$A_{\mathrm{phy}}^{\bar{+}}\left(\tilde{s}, u\right) = Q_{\mathrm{phy}}^{\pi_{\bar{\theta}^+}}\left(\tilde{s}, u\right) - V_{\mathrm{phy}}^{\pi_{\bar{\theta}^+}}\left(\tilde{s}\right). \tag{42}$$

Under the factorized decentralized policy,

$$\pi_\theta\left(u \mid \tilde{s}\right) = \prod_{i=1}^m \pi_{\theta_i}\left(u_i \mid \tilde{o}_i\right). \tag{43}$$

Assuming support coverage, the joint importance ratio and its clipped form are

$$\begin{aligned} \rho[n] &= \prod_{i=1}^m \frac{\pi_{\bar{\theta}_i^+}\left(u_i[n] \mid \tilde{o}_i[n]\right)}{\pi_{\theta_i^-}\left(u_i[n] \mid \tilde{o}_i[n]\right)}, \\ \bar{\rho}[n] &= \mathrm{clip}\left(\rho[n], 1-\epsilon_\rho, 1+\epsilon_\rho\right). \end{aligned} \tag{44}$$

The ratio corrects the action-distribution shift on stored samples but not the state-visitation shift.

A finite-sample first-order estimator corresponding to Theorem IV.1 is

$$\begin{aligned} \widehat{G}_{\theta\phi} &:= \frac{1}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} \nabla_{\theta^-} \log \pi_{\theta^-}\big(u[n] \mid \tilde{s}[n]\big)\left(\nabla_\phi \widehat{A}_\phi^-[n]\right)^\top, \\ \widehat{g}_{\mathrm{phy}}^{\bar{+}} &:= \frac{1}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} \mathrm{sg}\left(\bar{\rho}[n] \widehat{A}_{\mathrm{phy}}^{\bar{+}}[n]\right) \nabla_{\bar{\theta}^+} \log \pi_{\bar{\theta}^+}\big(u[n] \mid \tilde{s}[n]\big), \\ \widehat{g}_\phi^{\mathrm{meta}} &:= \alpha \widehat{G}_{\theta\phi}^\top \widehat{g}_{\mathrm{phy}}^{\bar{+}}, \end{aligned} \tag{45}$$

where $\mathrm{sg}(\cdot)$ denotes stop-gradient. The estimator remains approximate because of finite-step HATRL adaptation, critic approximation, clipped importance correction, and the fixed pre-update sampling distribution.

Motivated by Eqs. (38)–(45), the implemented teacher uses a physical-score-based outer proxy at active update boundaries. Let

$$\Delta J_k = J_{t_k}^{\mathrm{score}} - J_{t_{k-1}}^{\mathrm{score}}. \tag{46}$$

Its normalized form $\widehat{\Delta J}_k$ is obtained by exponential-moving-average centering and root-mean-square (RMS) standardization. The physical-score/teacher-output covariance is

$$\mathrm{Cov}_k(P, F_\phi) = \frac{1}{N_k} \sum_{n=1}^{N_k} \big(P_n - \bar{P}_k\big)\big(F_{\phi,n} - \bar{F}_{\phi,k}\big). \tag{47}$$

The teacher update uses

$$\begin{aligned}\mathcal{L}_{\text{teacher}} &= \mathcal{L}_{\text{base}} + \lambda_{\text{outer}} s_k \mathcal{L}_{\text{outer}},\\ \mathcal{L}_{\text{outer}} &\propto -\widehat{\Delta J}_k \,\mathrm{Cov}_k(P, F_\phi),\\ \widehat{g}^{\text{outer}}_{\phi,k} &= \nabla_\phi \mathcal{L}_{\text{teacher}},\end{aligned} \tag{48}$$

where $s_k \in [0,1]$ is the warm-up/ramp factor and the outer-proxy term is clipped during optimization. Thus, Eq. (45) provides the analytical motivation, whereas the implemented teacher uses the physical-score outer proxy above.

B. Conditional Local Ultimate Boundedness

The following analysis establishes conditional local ultimate boundedness of the closed-loop system under the deployed student policies.

Let $s_p, s_v, s_a > 0$ be characteristic error scales and define

$$\begin{aligned}e_p[n] &= \mathrm{col}\left\{e_{p,i}[n]\right\}_{i=1}^{m},\\ e_v[n] &= \mathrm{col}\left\{e_{v,i}[n]\right\}_{i=1}^{m},\\ e_a[n] &= \mathrm{col}\left\{e_{a,i}[n]\right\}_{i=1}^{m},\\ S_e &= \mathrm{diag}\left\{s_p I_m,\, s_v I_m,\, s_a I_m\right\},\\ \bar{e}[n] &= S_e^{-1}\,\mathrm{col}\left\{e_p[n], e_v[n], e_a[n]\right\}.\end{aligned} \tag{49}$$

Thus, $\bar{e}[n]$ is the dimensionless normalized tracking-error vector.

Similarly, define

$$\begin{aligned}\omega[n] &= \mathrm{col}\left\{f_a[n], f_p[n], w[n], \Delta_{\text{hold}}[n]\right\},\\ \bar{\omega}[n] &= S_\omega^{-1}\omega[n],\end{aligned} \tag{50}$$

where $f_a[n]$ and $f_p[n]$ are the aggregate FDI vectors, $w[n]$ is the aggregate disturbance vector, $\Delta_{\text{hold}}[n]$ collects packet-loss-induced holding errors, and S_ω is a positive diagonal scaling matrix.

Assumption 2 (Local dissipativity of the trained Student closed loop)**.** For the deployed student policies in the considered local operating domain, there exists $\bar{\omega} < \infty$ such that

$$\|\bar{\omega}[n]\| \le \bar{\omega}, \tag{51}$$

where $\bar{\omega}$ denotes the finite disturbance bound, and bounded packet-loss bursts or an equivalent communication admissibility condition ensure bounded $\Delta_{\text{hold}}[n]$. The deployed trajectory is assumed to remain in this local domain. Moreover, there exist a Lyapunov function $V(\bar{e})$ and constants $\underline{c} > 0$, $\bar{c} > 0$, $0 < \alpha_V < 1$, $c_\omega > 0$, and $c_{\text{cl}} > 0$ such that

$$\underline{c}\|\bar{e}[n]\|^2 \le V(\bar{e}[n]) \le \bar{c}\|\bar{e}[n]\|^2, \tag{52}$$

and

$$\begin{aligned}V(\bar{e}[n+1]) - V(\bar{e}[n]) \le{}& -\alpha_V V(\bar{e}[n])\\ &+ c_\omega\|\bar{\omega}[n]\|^2 + c_{\text{cl}}\varepsilon_{\text{cl}}^2,\end{aligned} \tag{53}$$

where $\varepsilon_{\text{cl}} \ge 0$ is a bounded deployment-time closed-loop residual arising from finite policy capacity and finite-step training and is fixed after training.

Theorem IV.2 (Conditional local ultimate boundedness)**.** *Under Assumption 2, any deployed trajectory remaining in the local operating domain satisfies*

$$\begin{aligned}V(\bar{e}[n]) \le{}& (1-\alpha_V)^n V(\bar{e}[0])\\ &+ \frac{1-(1-\alpha_V)^n}{\alpha_V}\left(c_\omega\bar{\omega}^2 + c_{\text{cl}}\varepsilon_{\text{cl}}^2\right).\end{aligned} \tag{54}$$

Consequently,

$$\limsup_{n\to\infty}\|\bar{e}[n]\| \le \bar{\delta}_{\text{UB}} := \sqrt{\frac{c_\omega\bar{\omega}^2 + c_{\text{cl}}\varepsilon_{\text{cl}}^2}{\alpha_V \underline{c}}}. \tag{55}$$

Therefore, the control objective in Eq. (12) *is satisfied whenever*

$$\delta_q \ge s_q \bar{\delta}_{\text{UB}}, \qquad q \in \{p, v, a\}, \tag{56}$$

where $s_q \in \{s_p, s_v, s_a\}$ *corresponds to* $q \in \{p, v, a\}$.

Proof. From Eq. (53) and $\|\bar{\omega}[n]\| \le \bar{\omega}$,

$$\begin{aligned}V(\bar{e}[n+1]) \le{}& (1-\alpha_V)V(\bar{e}[n])\\ &+ c_\omega\bar{\omega}^2 + c_{\text{cl}}\varepsilon_{\text{cl}}^2.\end{aligned} \tag{57}$$

Iterating this recursion yields Eq. (54). Taking the upper limit and using the lower bound in Eq. (52) yields Eq. (55). Moreover, by the definition of the normalized error vector,

$$|e_{q,i}[n]| \le s_q\|\bar{e}[n]\|, \qquad q \in \{p, v, a\}. \tag{58}$$

Therefore, condition (56) implies the componentwise objective in Eq. (12). □

V. Simulation Results and Analysis

This section evaluates the proposed MGRS framework under hybrid DoS–FDI network attacks through comparative simulations and attack-response analyses.

A. Simulation Setup and Adversarial Evaluation Protocol

The longitudinal platooning task is evaluated in a physics-based planar vehicle environment implemented in NVIDIA Isaac Sim using Isaac Lab and the PhysX engine. The platoon comprises one leader and four followers modeled using the WHEELTEC `senior_4wd_bs_robot` unified robot description format (URDF) model. Under CTDE, each actor generates a scalar longitudinal command from its attacked local observation, while the centralized critic uses the corrupted joint observation. The leader follows a predefined longitudinal reference, and the followers regulate their longitudinal spacing along prescribed straight paths. Figure 2 shows representative views of the simulation environment, and Table I summarizes the implementation, timing, training, and agent-interface settings.

During training, CA-GAN generates state-conditioned hybrid DoS–FDI attacks under progressively increased attack budgets. Two evaluation protocols are considered. First, MGRS is evaluated across nine progressively increased CA-GAN attack-budget levels to characterize its response to attack intensity. Second, the trained policies are frozen and evaluated under the highest-intensity CA-GAN attack condition for cross-method comparison.

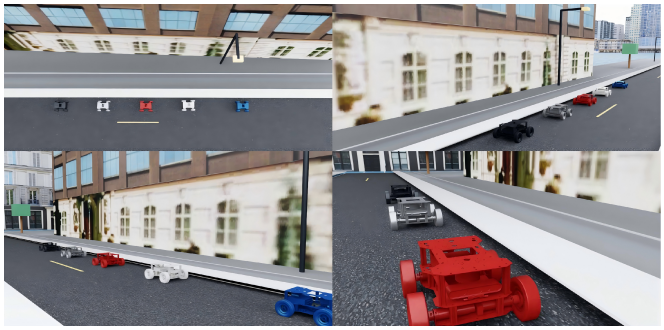


Fig. 2. Multi-view rendering of the five-vehicle platoon in NVIDIA Isaac Sim.

TABLE I
KEY SIMULATION AND CONTROL PARAMETERS

Category	Parameter	Setting
Implementation	Simulator / framework / physics / platform	Isaac Sim 5.1.0 / Isaac Lab 0.53.0 / PhysX; Ubuntu 22.04.5; RTX 3080 10 GB
Vehicle platoon	URDF / formation	WHEELTEC model; 1 leader + 4 followers
Timing and training	Physics / control intervals; environments / episode / seed	0.005 s / 0.020 s; 64 / 20 s / 42
Agent interface	Local observation / critic input / policy action	11 / 55 / scalar longitudinal command
Actuator interface	Command mapping	Deterministic mapping

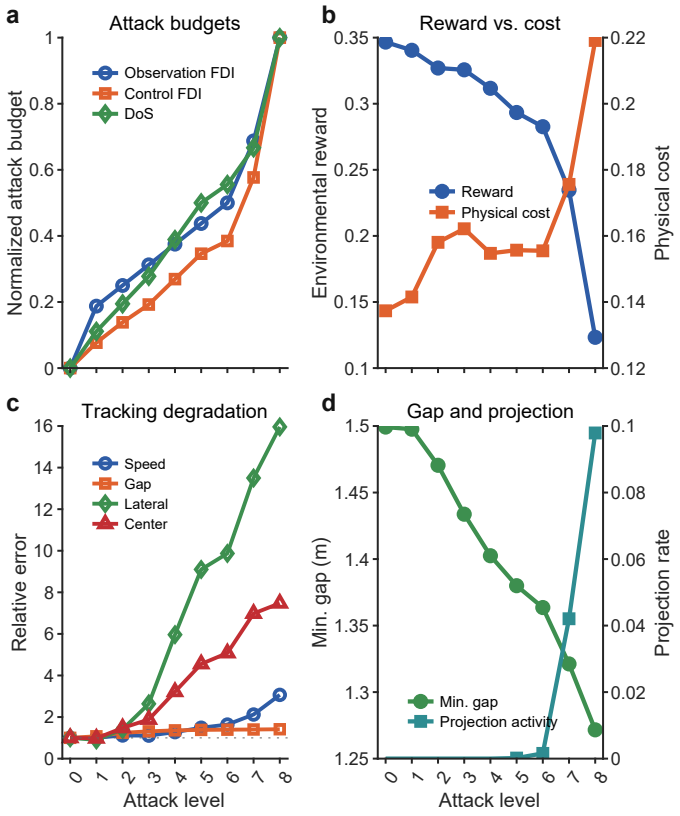


Fig. 3. MGRS performance across CA-GAN attack-budget levels: (a) attack budgets, (b) reward and physical cost, (c) tracking errors, and (d) minimum gap and projection activity.

At the highest attack level, the CA-GAN acts on all vehicles, local observation channels, and longitudinal command channels. The observation- and control-channel FDI outputs are bounded by the prescribed budgets ε_p and ε_a, respectively, while the component-level DoS probabilities satisfy

$$0 \le q_{p,i}^{(j)}, q_{a,i}^{(k)} \le \rho_{\mathrm{DoS}}^{\max}, \qquad \rho_{\mathrm{DoS}}^{\max} = 0.18. \tag{59}$$

Packet losses retain the most recent valid values through ZOH. The action projection is retained during this comparison; hence, Figs. 5–6 and Table II report projection-assisted frozen-policy performance.

B. Simulation Performance Evaluation and Analysis

The influence of progressively intensified hybrid attacks is first examined in Figs. 3 and 4. As the CA-GAN attack budgets increase from the zero-attack to the highest-intensity condition, the mean environmental reward decreases by 64.4%, while the tracking and geometric-deviation metrics increase overall and the mean minimum pair gap decreases from 1.4992 m to 1.2716 m. The fine-grained results further show that the resulting degradation is nonuniform across the platoon. At the highest attack level, vehicle 3 exhibits the largest centerline geometric deviation of 0.196 m, while pairs 2 and 3 exhibit the largest pairwise lateral deviations of 0.227 m and 0.206 m, respectively. The stage-local reward decreases from the beginning to the end at all nine attack levels, with the largest reduction of 54.5% occurring under the highest-intensity attack. These results indicate that the influence of hybrid attacks becomes increasingly pronounced with the attack budget and varies across individual vehicles and inter-vehicle pairs.

The compared methods are MAPPO [23], HAPPO [24], MGRS, MAPPO-S [33], Heterogeneous-Agent Advantage Actor-Critic (HAA2C) [32], and HATRPO [24], with HATRL [32] serving as the lower-level optimizer of MGRS. MAPPO-S denotes the shared-policy MAPPO configuration. All compared methods are evaluated under the same highest-intensity CA-GAN attack condition, with state-conditioned attack realizations generated from the current platoon state.

Under this most severe attack condition, Fig. 5 compares the cumulative returns of the six methods. MGRS achieves the highest final and last-three-checkpoint mean cumulative returns of 18.287×10^3 and 18.261×10^3, exceeding HAPPO by 3.83% and 3.98%, respectively. These results show that MGRS maintains the highest cumulative-return performance among the compared methods under the most severe hybrid-attack condition considered.

The corresponding physical-control performance is further evaluated in Table II and Fig. 6. MGRS achieves the lowest speed and spacing mean absolute errors (MAEs), the lowest centerline geometric deviation, and the largest mean minimum pair gap. Relative to HAPPO, the speed and spacing MAEs decrease by 18.24% and 6.50%, respectively, while the cen-

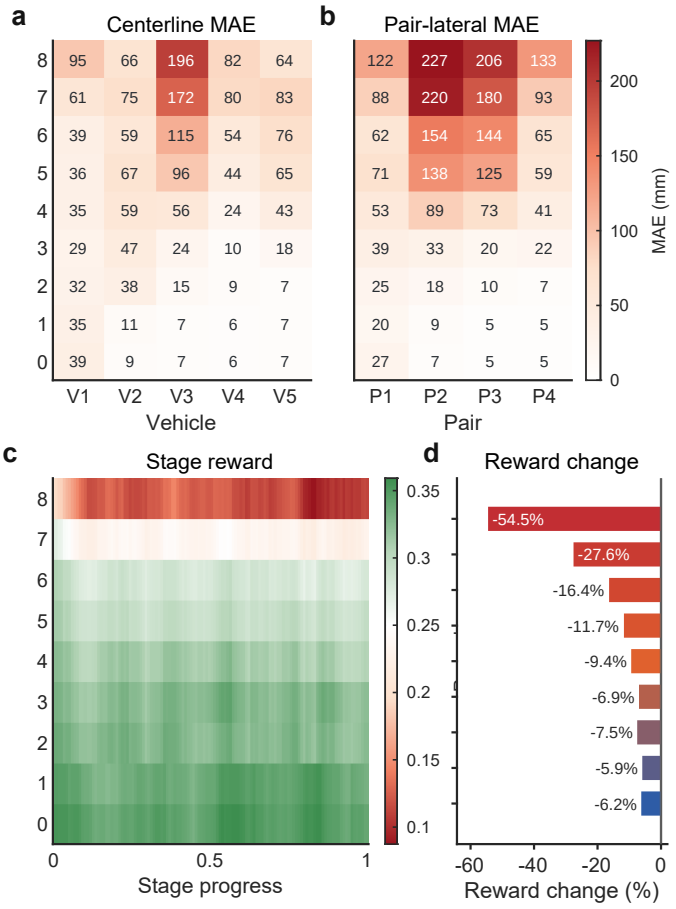


Fig. 4. Fine-grained CA-GAN attack-level diagnostics: (a) vehicle-level centerline geometric deviation, (b) pairwise lateral geometric deviation, (c) stage-local reward, and (d) start-to-end reward change.

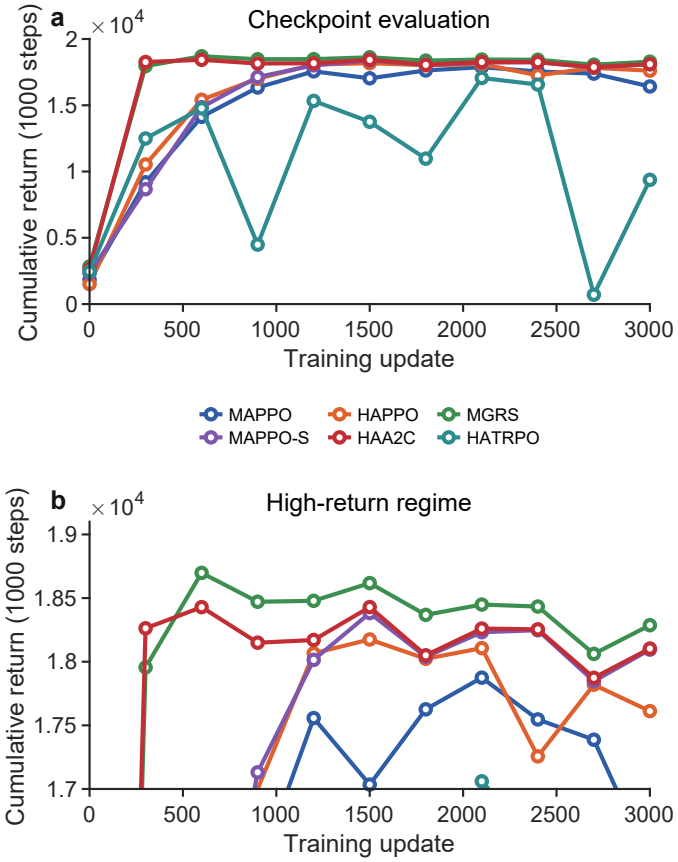


Fig. 5. Cumulative returns of the six compared methods under the highest-intensity CA-GAN attack condition: (a) all checkpoints and (b) enlarged high-return regime.

terline geometric deviation decreases by 69.36%. MGRS also reduces the lateral geometric deviation to 0.0088 m, close to the best value of 0.0083 m achieved by MAPPO-S. During 3000 meta-training updates, the Reward Teacher is activated at 1500 update boundaries. As shown in Fig. 7, the shaping signal evolves with the normalized physical-score change and the physical-score/Teacher-output covariance, showing that reward shaping responds to the physical-performance evolution during training. Together with the quantitative control improvements above, these results support that the reward-adjustment mechanism reduces the influence of attack-corrupted environmental rewards on policy optimization, thereby mitigating attack-induced reward misalignment and improving platoon-control performance under hybrid attacks.

VI. Physical-Platform Experimental Validation

This section evaluates the deployment feasibility of the trained decentralized student policies through a simultaneous four-follower physical-platoon experiment under CA-GAN-generated hybrid DoS–FDI attacks on the downlink control channel.

A. Experimental Platform and Online Implementation

Figure 8 shows the experimental platform and hardware–software architecture. The platform comprises one virtual leader and four WheelTec05 physical followers. Each follower integrates an NVIDIA Jetson TX2 and a microcontroller connected via USART, while a NOKOV motion-capture (MoCap) system provides planar pose measurements and the ground station communicates with the vehicles through the Robot Operating System (ROS) over Wi-Fi. Policy inference, reference generation, CA-GAN attack injection, and data recording are implemented through the Python/Isaac Lab–ROS pipeline shown in Fig. 8(b).

Each follower executes an independently parameterized student policy using only its local attacked observation. Uncompromised physical measurements are recorded for evaluation but are not supplied to the student action-selection rule. The Reward Teacher and training-time action projection are removed during physical deployment, while an independent emergency-stop mechanism is retained only for hardware protection.

For physical evaluation, the virtual leader and four physical followers operate simultaneously on a circular track due to the limited experimental area. The virtual leader specifies the longitudinal reference on a path with a radius of 1.28 m, while the followers regulate their assigned phase and spacing offsets during the approximately 40-s experiment, with the four decentralized student policies running concurrently. CA-GAN-generated hybrid DoS–FDI attacks are applied through the downlink attack interface, and the light-red bands in

TABLE II
PROJECTION-ASSISTED PERFORMANCE UNDER THE HIGHEST-INTENSITY CA-GAN ATTACK CONDITION

Configuration	Final ret. ↑	Late ret. ↑	Speed ↓	Spacing ↓	Lat. geom. ↓	Ctr. geom. ↓	Min. gap ↑
MAPPO [23]	16.418	17.117	0.0768	0.2395	0.0707	0.0348	1.3687
HAPPO [24]	17.612	17.562	0.0636	0.1830	0.0173	0.0297	1.4987
MGRS	**18.287**	**18.261**	**0.0520**	**0.1711**	0.0088	**0.0091**	**1.5114**
MAPPO-S [33]	18.095	18.063	0.0659	0.2114	**0.0083**	0.0136	1.4994
HAA2C [32]	18.103	18.077	0.0660	0.2079	0.0155	0.0177	1.4991
HATRPO [24]	9.376	8.881	0.1352	0.2635	0.4347	0.2999	1.3651

Returns are divided by 10^3. "Late ret." is the mean of the final three checkpoints. Speed is in m/s; spacing, lateral geometric deviation, centerline geometric deviation, and minimum-gap values are in m. Bold and underlining indicate the best and second-best values, respectively.

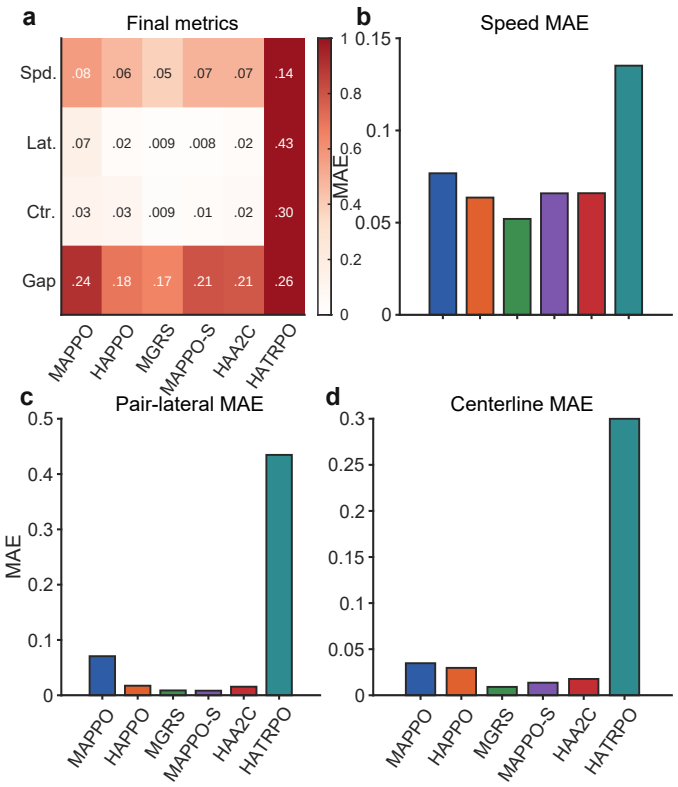


Fig. 6. Final-checkpoint metrics under the highest-intensity CA-GAN attack condition: (a) normalized overview, (b) speed MAE, and (c)–(d) geometric-deviation indicators.

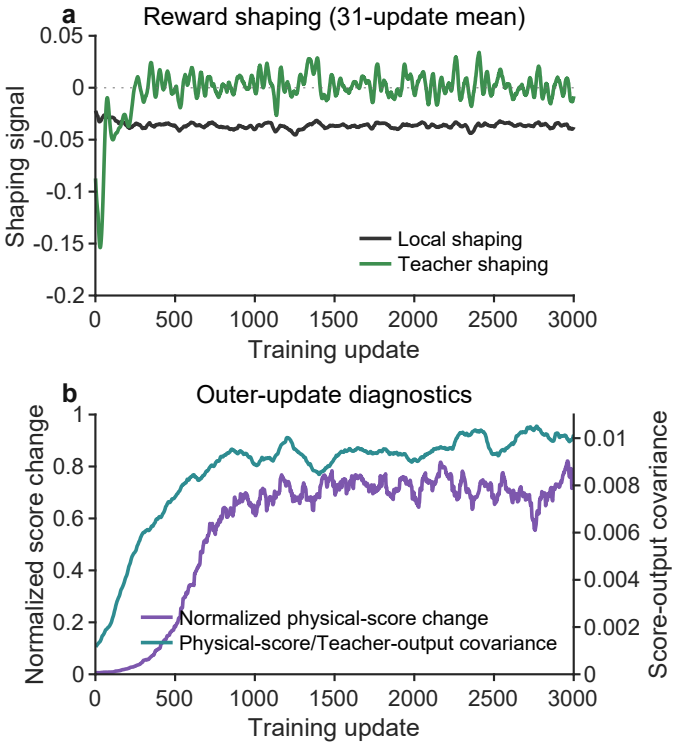


Fig. 7. Reward-Teacher training and outer-update diagnostics: (a) reward-shaping signals and (b) normalized physical-score change and physical-score/Teacher-output covariance over active Teacher updates.

Fig. 9(d)–(f) indicate the intervals with the highest applied attack intensity. The simultaneous protocol retains inter-vehicle coupling and shared wireless communication during physical evaluation.

B. Physical Platoon-Control Results

Figures 9(a) and 9(b) show the measured trajectories and longitudinal positions of the four physical followers relative to the virtual-leader reference. The followers preserve their relative motion order throughout the experiment. As shown in Fig. 9(c), the radial root-mean-square error (RMSE), final anchor error, and absolute radius bias remain at the centimeter scale for all four followers, supporting the physical deployment feasibility of the trained student policies. Here, the final anchor error denotes the terminal distance to the assigned reference anchor, and the absolute radius bias is the difference in magnitude between the fitted and reference radii.

Figures 9(d)–(f) show the longitudinal position, velocity, and acceleration tracking errors under CA-GAN-generated hybrid DoS–FDI attacks on the downlink control channel. The errors exhibit transient fluctuations during the highlighted high-intensity attack intervals without sustained growth. The remaining deviations reflect the combined effects of physical vehicle dynamics, sensing, communication, and inter-vehicle coupling.Overall, the experimental results show that, under the proposed control method, the four followers maintain continuous coordinated tracking under hybrid attacks, with centimeter-scale geometric deviations and no sustained growth in longitudinal tracking errors, validating the deployment feasibility and attack-resistant control performance of the proposed method in physical vehicle platoons.

VII. CONCLUSION

A hierarchical MGRS framework was developed for security control of vehicle platoons under hybrid DoS–FDI attacks.

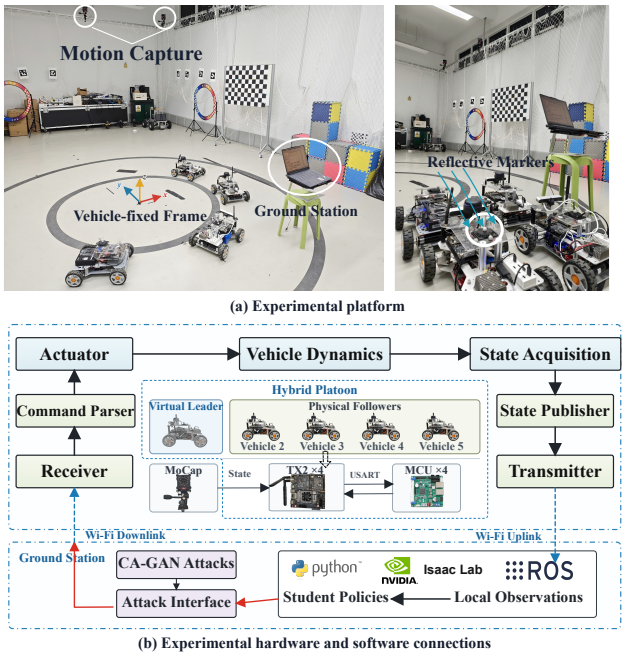


Fig. 8. Physical experimental platform and hardware–software architecture.

The framework combines CA-GAN-based adversarial training, decentralized HATRL student policies, and a physics-aware Reward Teacher to correct attack-induced reward misalignment under hybrid DoS–FDI attacks. Under the highest-intensity CA-GAN attack condition, MGRS achieves a higher cumulative return and lower speed and spacing MAEs than the compared methods, together with a lower centerline geometric deviation. A four-follower physical-platform experiment further supports the deployment feasibility of the trained policies under CA-GAN-generated hybrid DoS–FDI attacks on the downlink control channel, with centimeter-scale radial geometric deviations. Future work will investigate reward correction with reduced reliance on privileged information and formal verification of deployed learned controllers.

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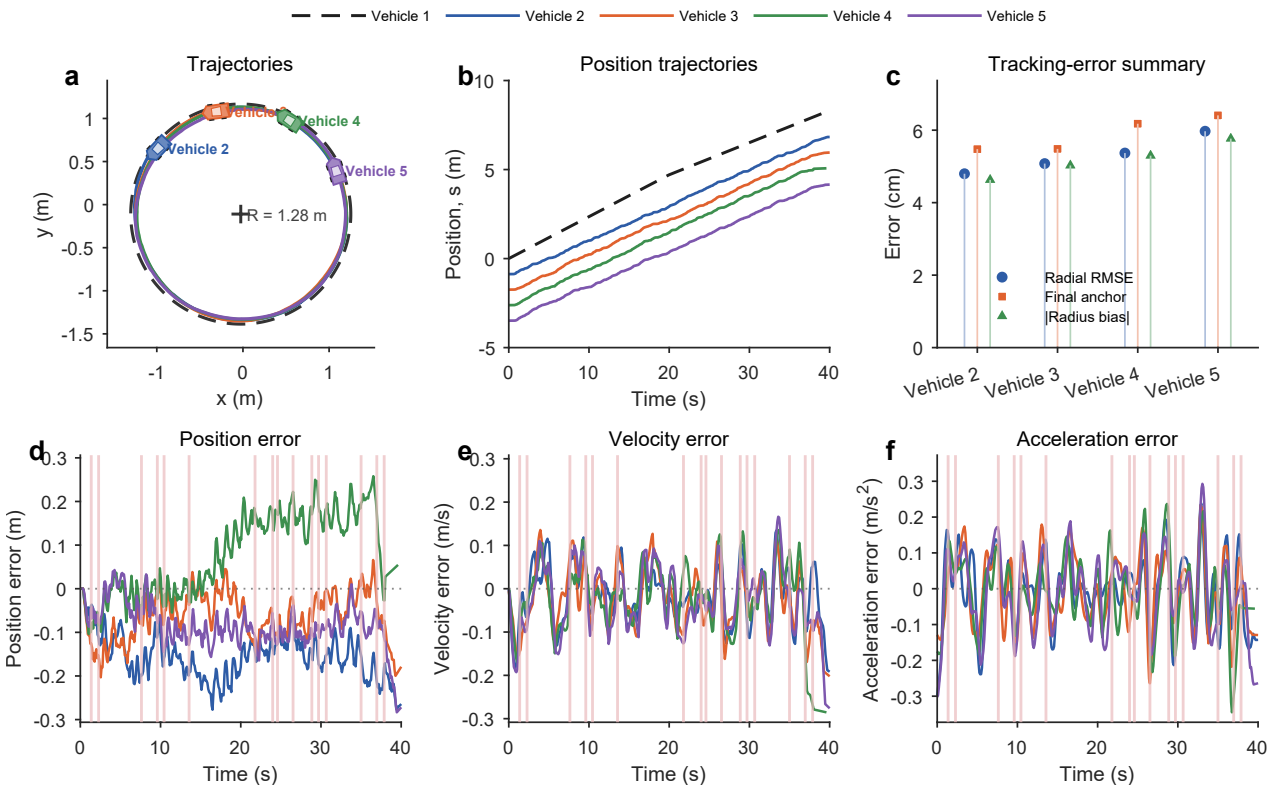


Fig. 9. Physical-platform results: (a) vehicle paths, (b) longitudinal positions, (c) radial geometric-deviation metrics, and (d)–(f) longitudinal position, velocity, and acceleration errors.

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